

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

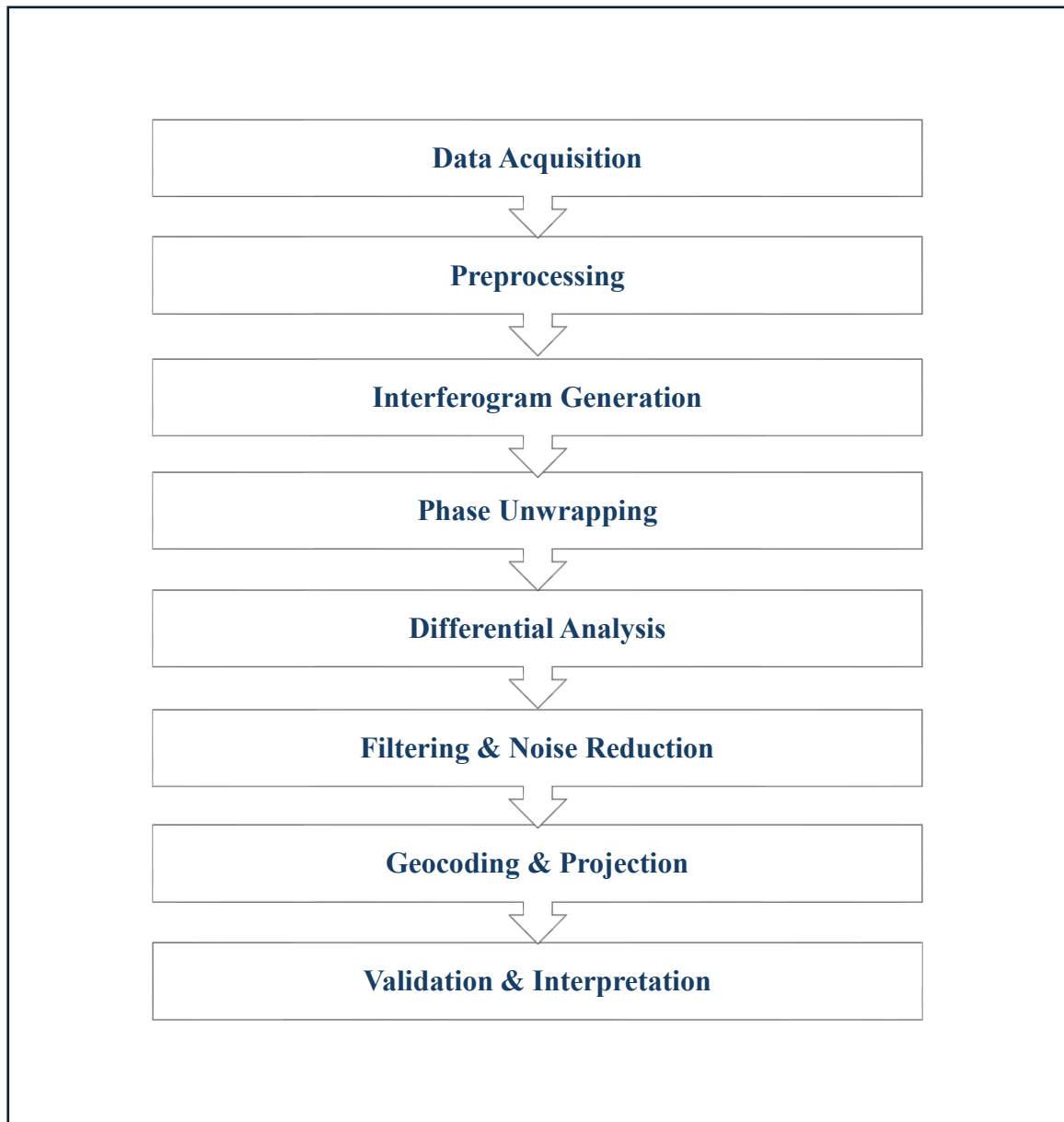
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 938+250

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

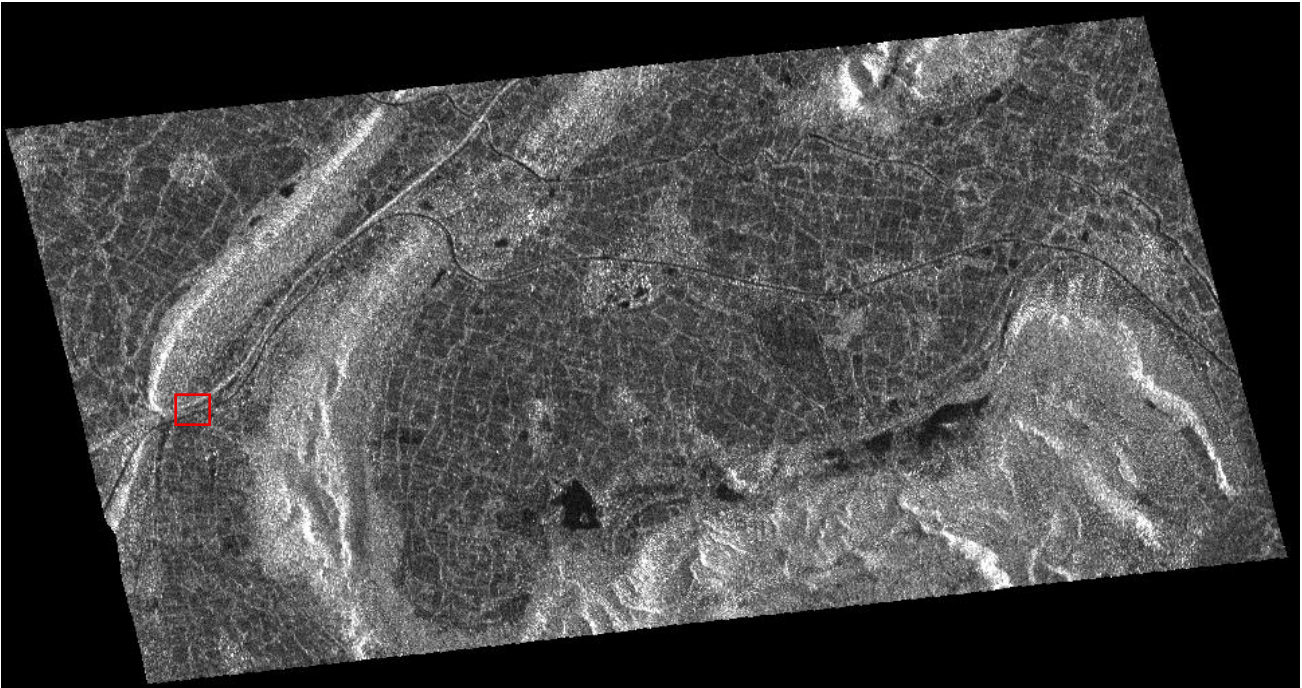
Flight Direction - DESCENDING

Polarization – VV+VH

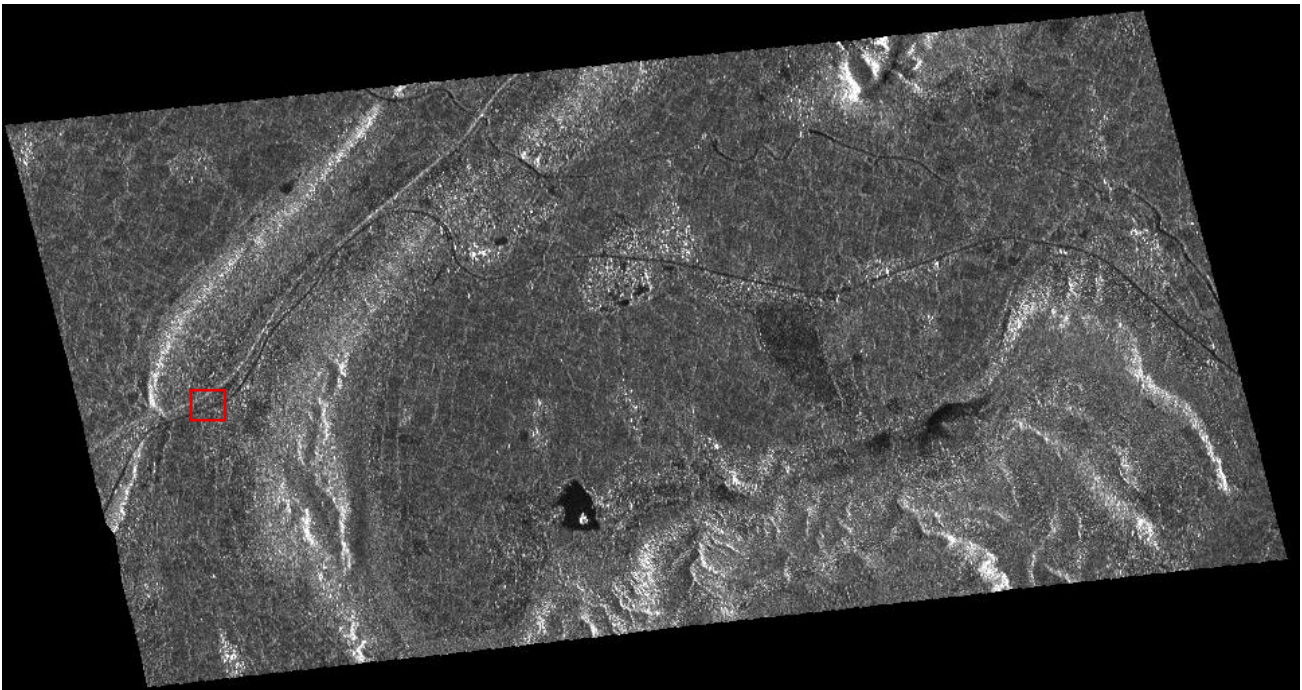
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

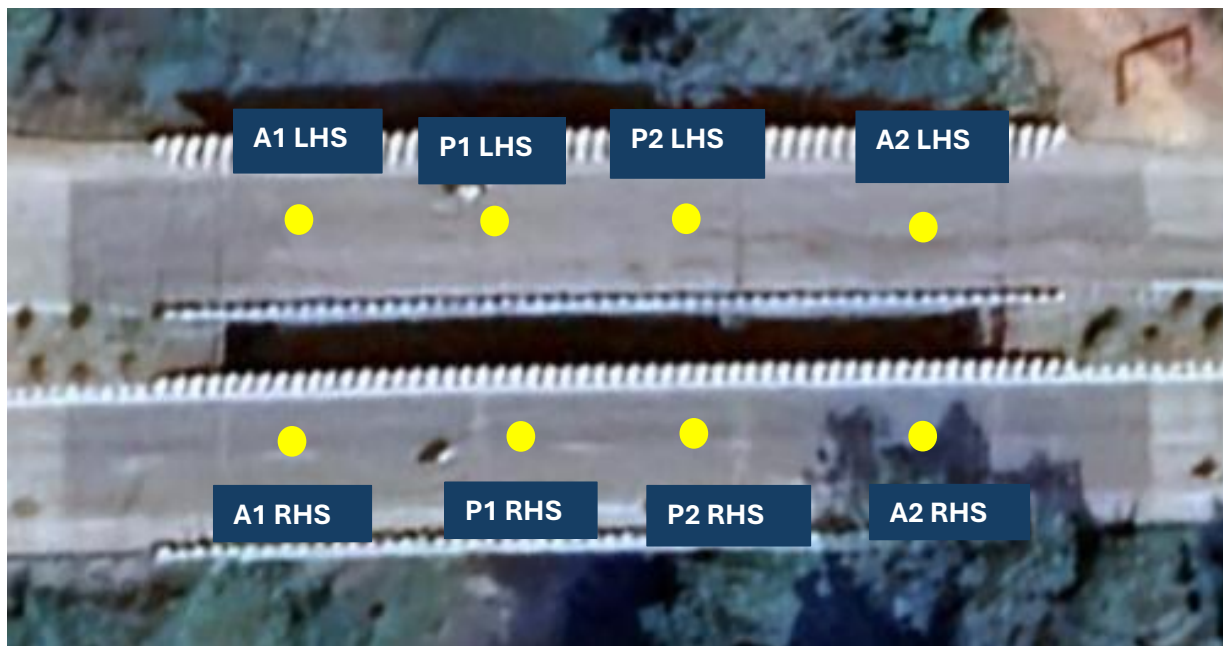


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

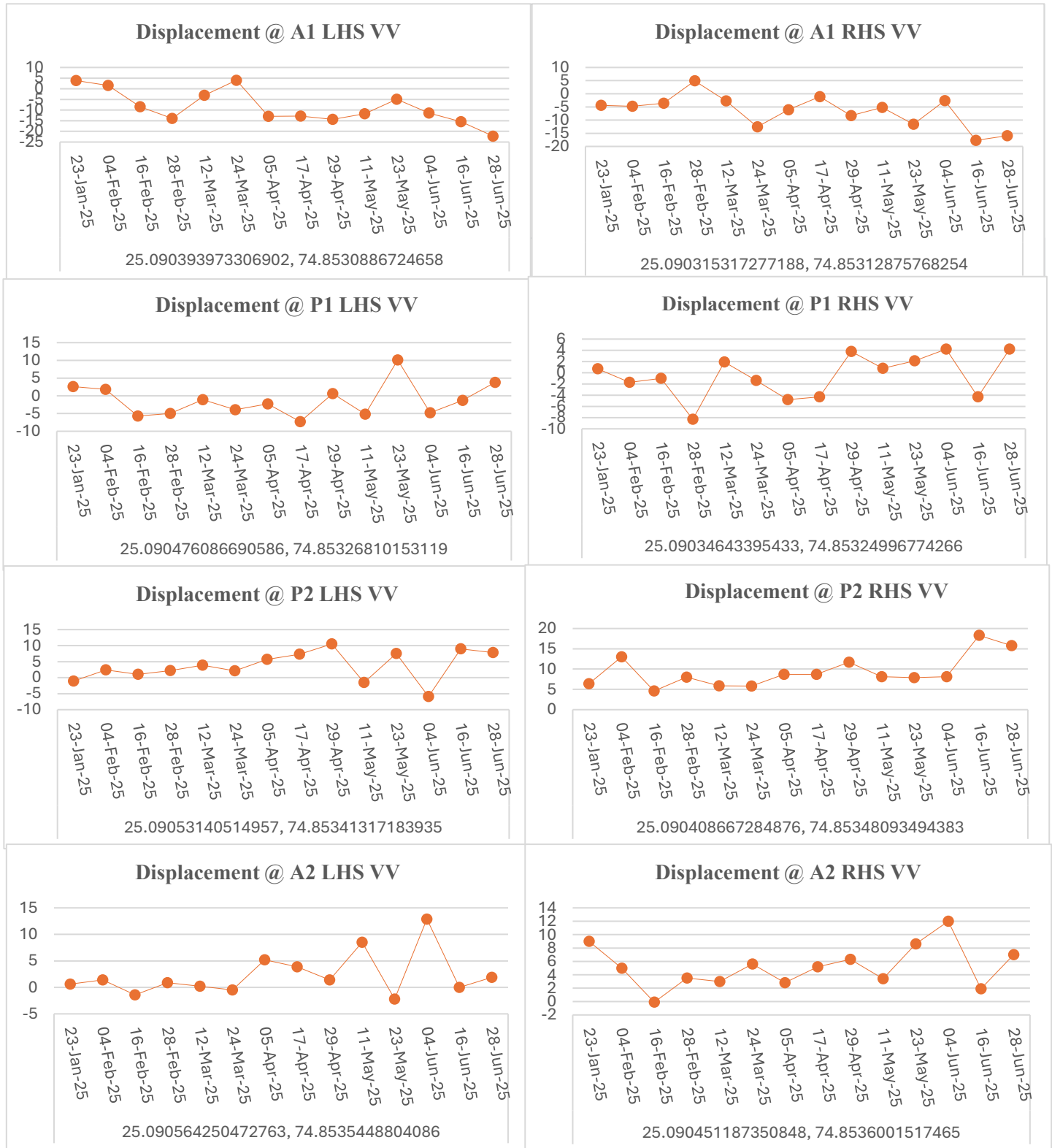
MNB – CH-938+250

Length – 56.25M



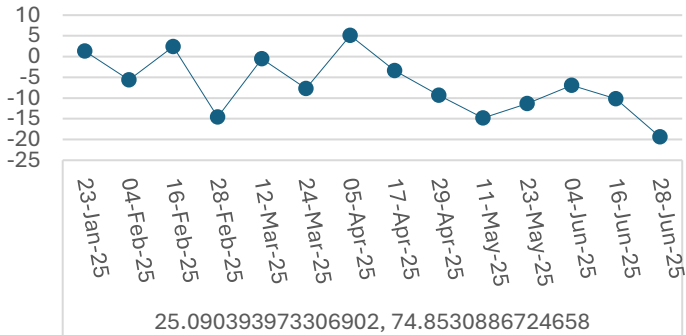
Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
P1	Pier 1		
P2	Pier 2		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

3.1 VV Displacement From 23 Jan 2025 to 28 June 2025

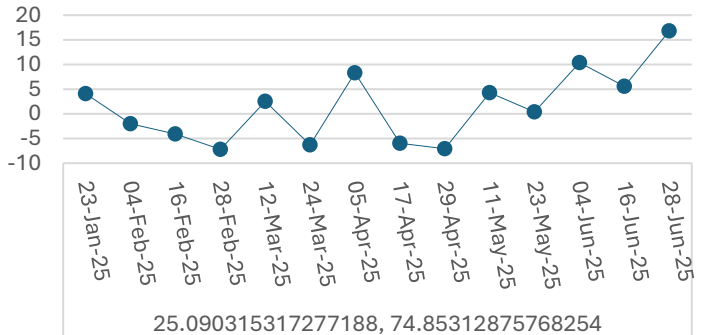


3.2 VH Displacement From 23 Jan 2025 to 28 June 2025

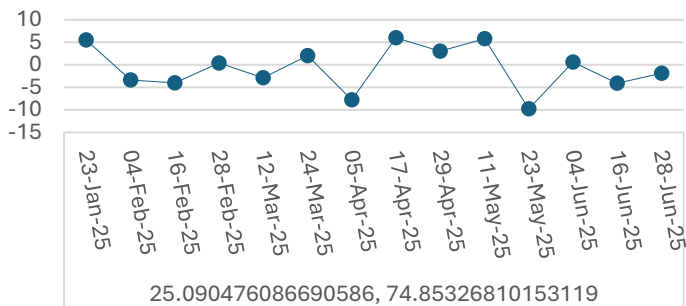
Displacement @ A1 LHS VH



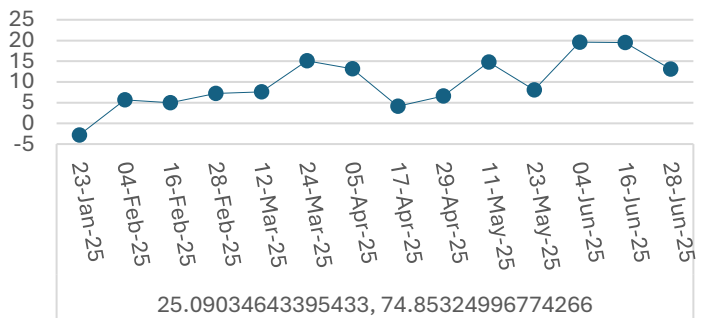
Displacement @ A1 RHS VH



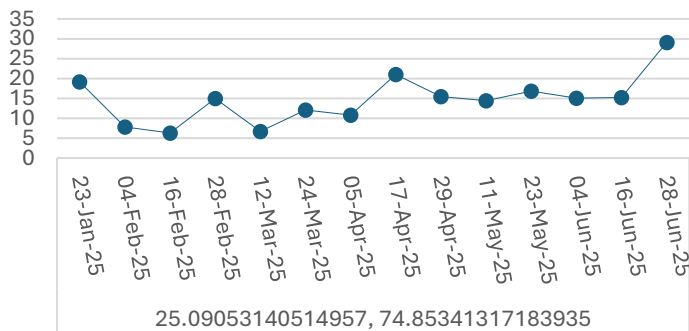
Displacement @ P1 LHS VH



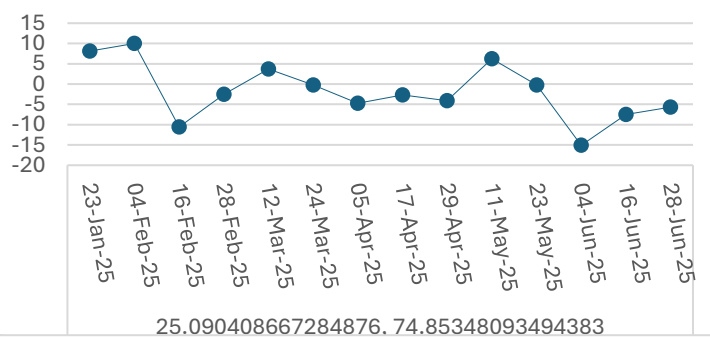
Displacement @ P1 RHS VH



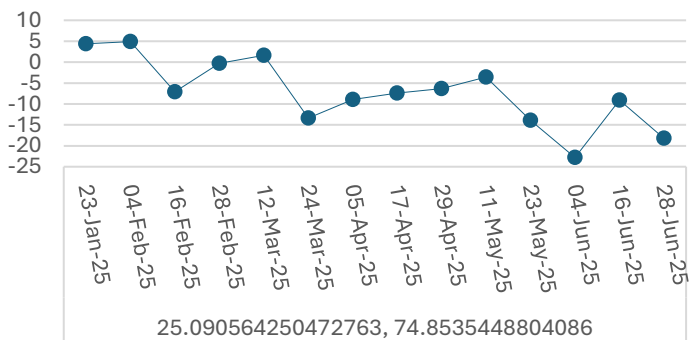
Displacement @ P2 LHS VH



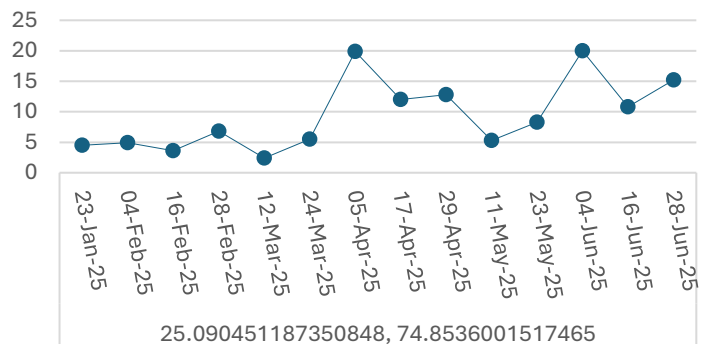
Displacement @ P2 RHS VH



Displacement @ A2 LHS VH



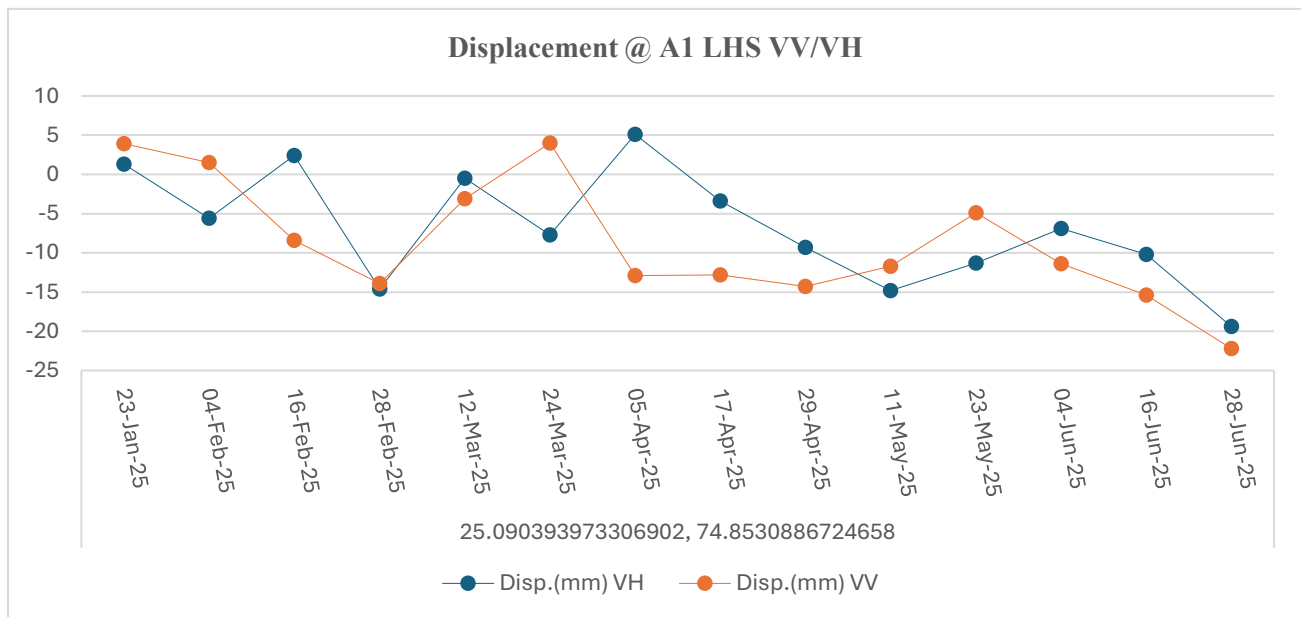
Displacement @ A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.090393973306902, 74.8530886724658)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows progressively increasing negative values through the monitoring period.
- **Horizontal displacement (VH)** displays moderate variability, ranging from +5 mm to -18 mm.

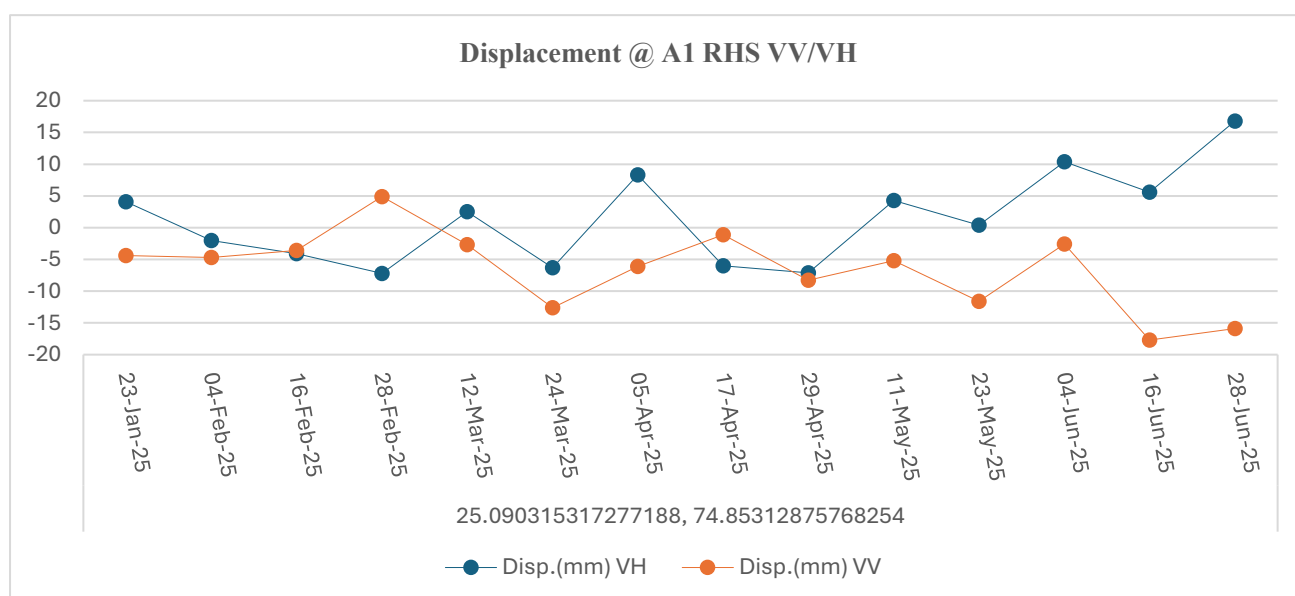
Critical Events & Anomalies:

- **March 2025:** Both VH and VV show a sharp drop (around -10 mm to -15 mm), marking a distinct downward event affecting both vertical and horizontal displacement.
- **April–May 2025:** VV remains consistently negative (between -10 mm to -18 mm), indicating sustained downward settlement.
- **June 2025:** Both VH and VV record some of the most negative values of the entire period (approaching -20 mm), indicating the continuation of structural movement and potential instability at this abutment location.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.090315317277188, 74.85312875768254)



Key Observations:

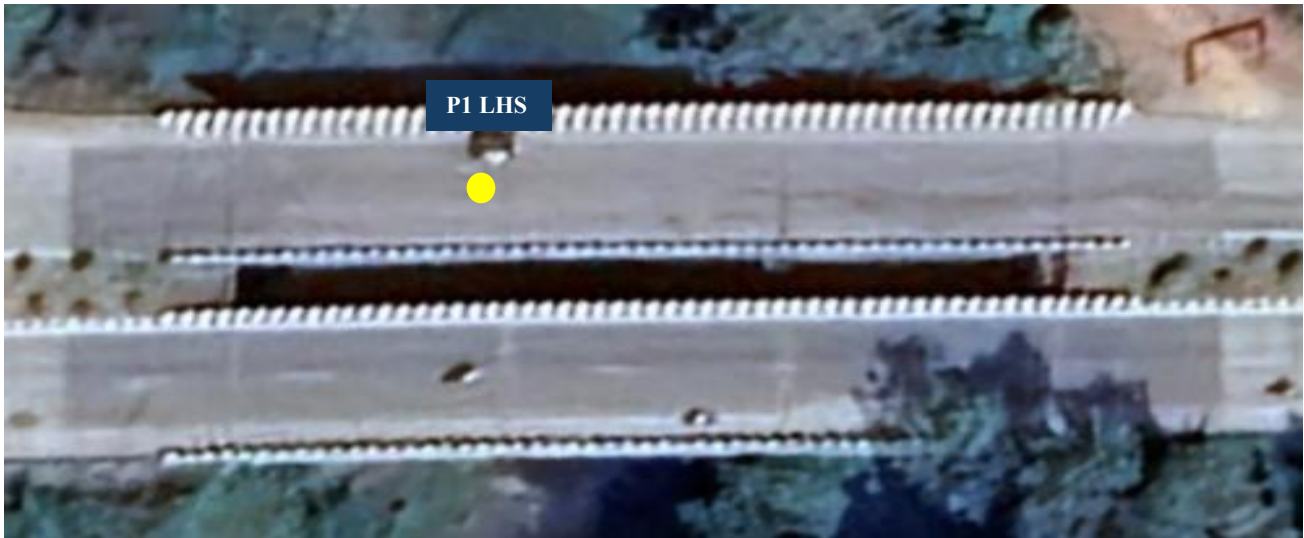
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH shows alternating upward and downward movement in February–April 2025, followed by a consistent upward trend after May 2025.
- **Vertical displacement (VV)** shows moderate variation throughout the monitoring period, mostly ranging between 0 mm and –15 mm.

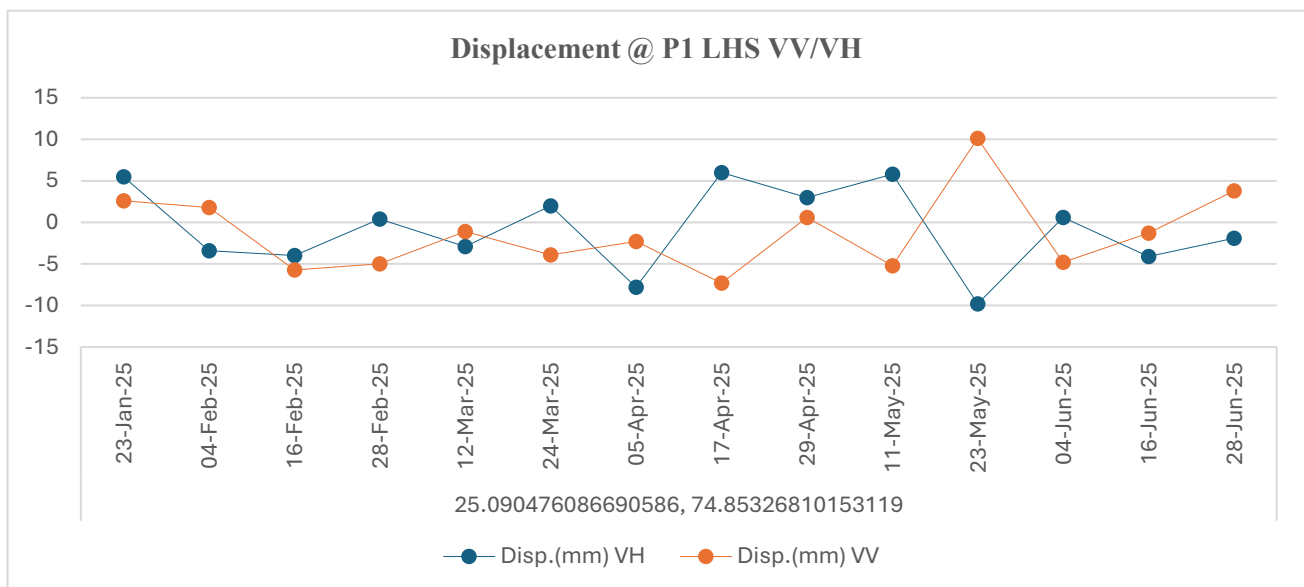
Critical Events & Anomalies:

- **March 2025:** VV records a sharp drop to around –12 mm, indicating a short-duration settlement event. VH also shows a corresponding dip, reflecting simultaneous vertical and lateral movement.
- **April–May 2025:** VV remains consistently negative (between –10 mm to –15 mm), indicating sustained downward displacement, while VH begins transitioning upward, showing divergence between vertical and horizontal movement.
- **June 2025:** VH increases sharply toward +15 to +18 mm, marking a significant lateral shift.

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (25.090476086690586, 74.85326810153119)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows moderate variability, generally ranging between **+10 mm** and **-10 mm**.
- **Horizontal Displacement (VH)** also remains within a moderate range, mostly between **+6 mm** and **-10 mm**. VH shows fluctuating behaviour but stays relatively centered around zero. One significant dip to approximately **-10 mm** occurs in late May, followed by partial recovery.

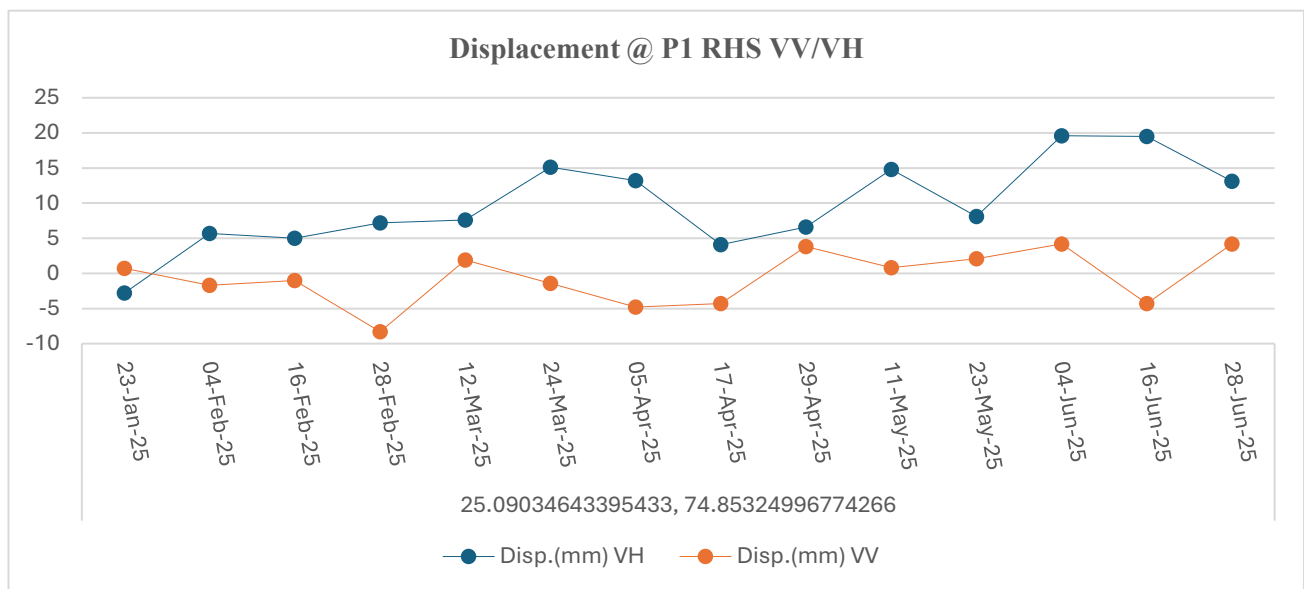
Critical Events & Anomalies:

- **May 2025:** A distinct divergence occurs—VV rises sharply to around **+12 mm**, while VH simultaneously drops to about **-10 mm**, indicating conflicting vertical uplift and lateral movement at the same time.
- **April 2025:** VH shows a notable upward movement (**~+6 mm**), marking one of its positive peaks, while VV remains relatively stable.
- Overall, both VV and VH exhibit **frequent short-term fluctuations**, but no extreme long-term progressive movement is observed.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (25.09034643395433, 74.85324996774266)



Key Observations:

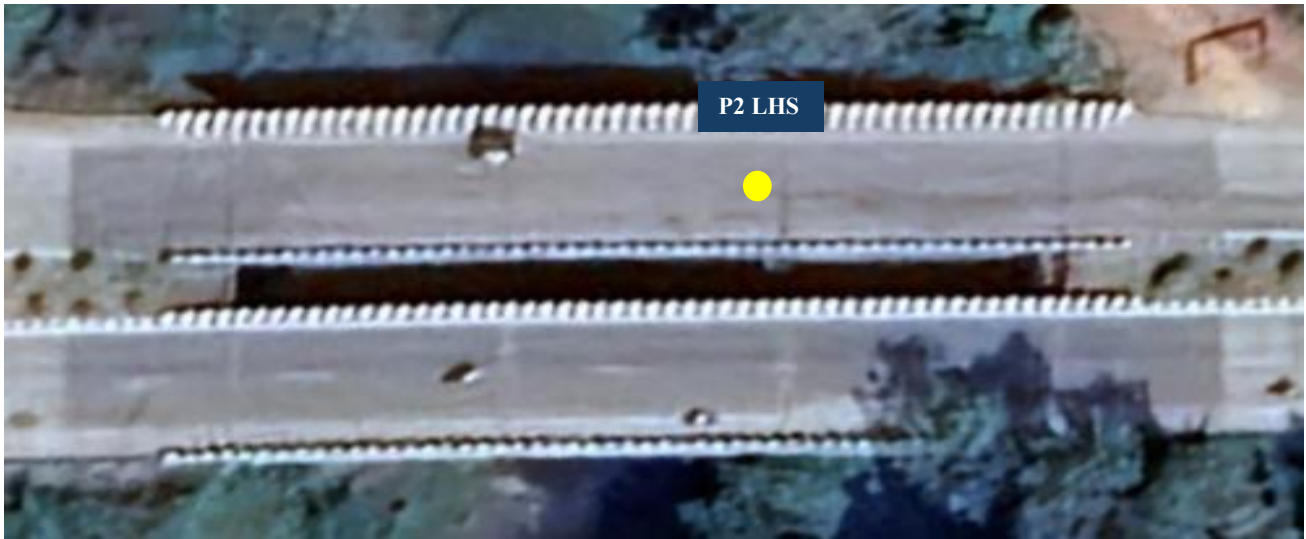
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** remains relatively stable with values mostly between **-8 mm and +5 mm**.
- **Horizontal Displacement (VH)** shows a clearer upward trend over time, starting near **0 to +5 mm** in January and gradually rising to **+15–20 mm** by late May and June.

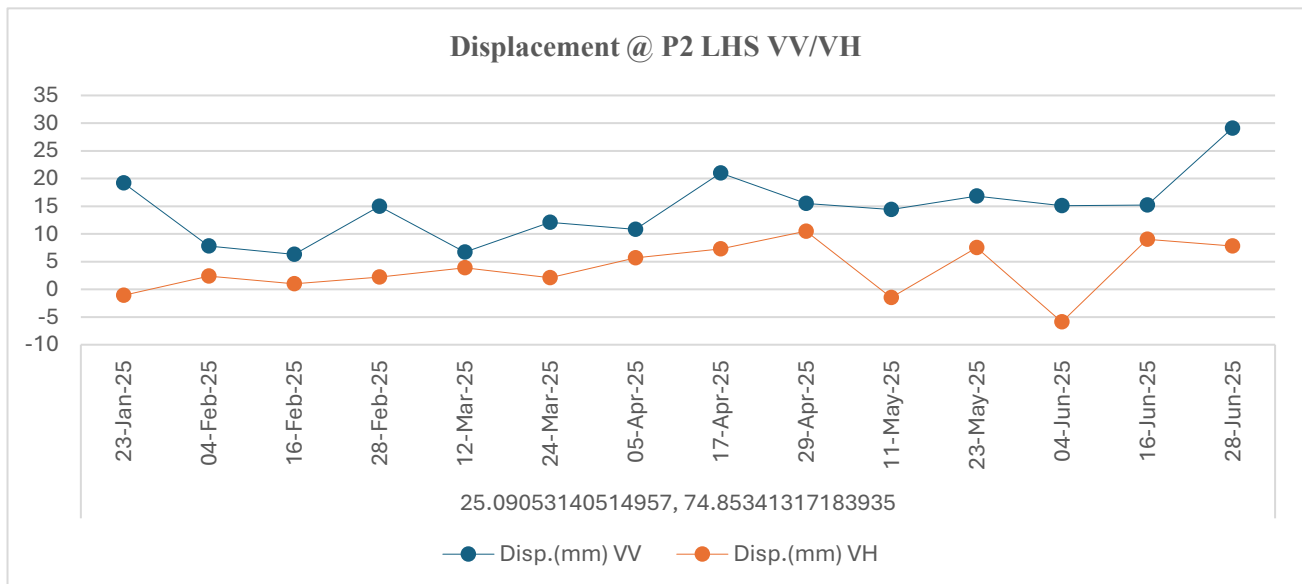
Critical Events & Anomalies:

- **March–April 2025:** VH begins a noticeable upward progression, crossing **+10 mm**, while VV remains stable showing separation between vertical and horizontal behaviours.
- **May 2025:** VH reaches its highest levels (**+20 mm**), indicating peak lateral movement. Meanwhile, VV stays within its typical moderate range (**-2 to +2 mm**), showing no corresponding vertical deformation.
- Overall, VH demonstrates **consistent lateral drift**, while VV displays **mild but stable variability**, suggesting lateral deformation at P1 RHS without significant vertical instability.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (25.09053140514957, 74.85341317183935)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** remains relatively **stable and low-magnitude**, generally varying between **−5 mm to +10 mm**. Aside from a short negative dip in late May, the horizontal behaviour remains mild and does not show any long-term drift.
- **Vertical Displacement (VV)** shows a **clear increasing trend** across the monitoring period. Values begin around **+20 mm** in January, fluctuate moderately through the mid-period, and rise sharply to approximately **+30–32 mm** by late June. This indicates a gradual upward vertical movement at P2 LHS.

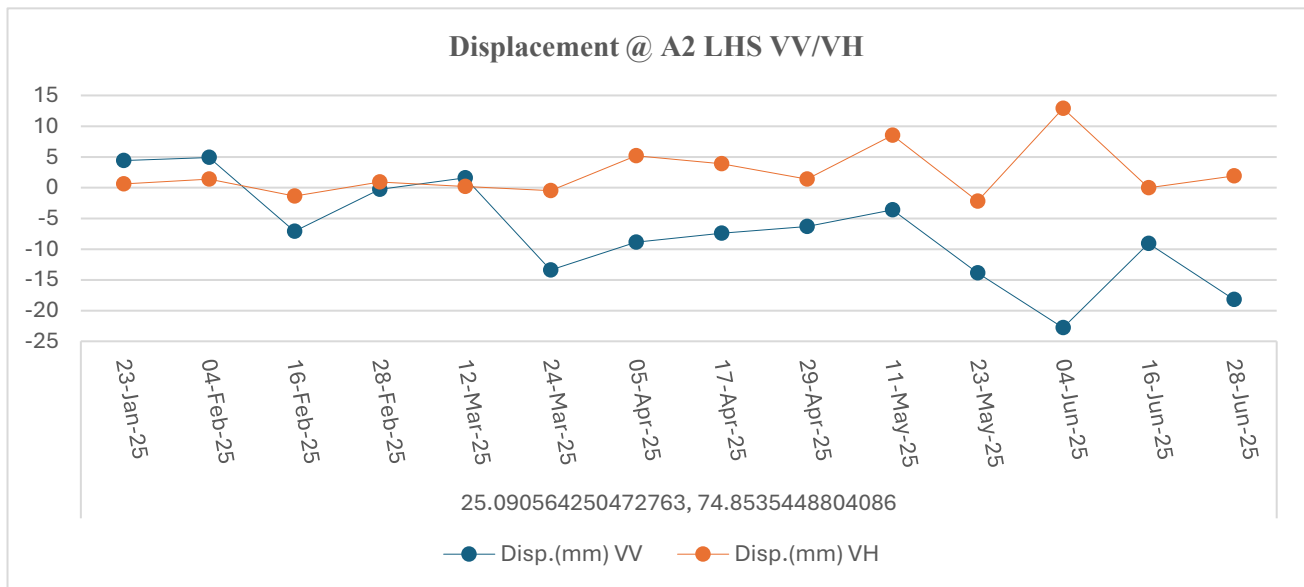
Critical Events & Anomalies:

- **May 2025:** VH shows a temporary negative shift (**~−5 mm**), but it promptly returns to the positive range, indicating a short-duration horizontal deviation rather than a structural drift.
- **June 2025:** VV peaks at over **+30 mm**, representing the maximum vertical displacement recorded in the dataset



- **March 2025:** VV drops sharply to around **-15 mm**, representing the largest downward vertical movement observed.

- **June 2025:** VH reaches its peak (around +17 to +18 mm), indicating the strongest horizontal displacement recorded in the dataset



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows higher variability with multiple downward movements, ranging roughly from +5 mm to –20 mm.
- **Horizontal Displacement (VH)** remains mostly positive and stable, generally between 0 mm and +10 mm, except for a few small negative fluctuations. VH shows an upward shift during mid-May and early June before stabilizing again.

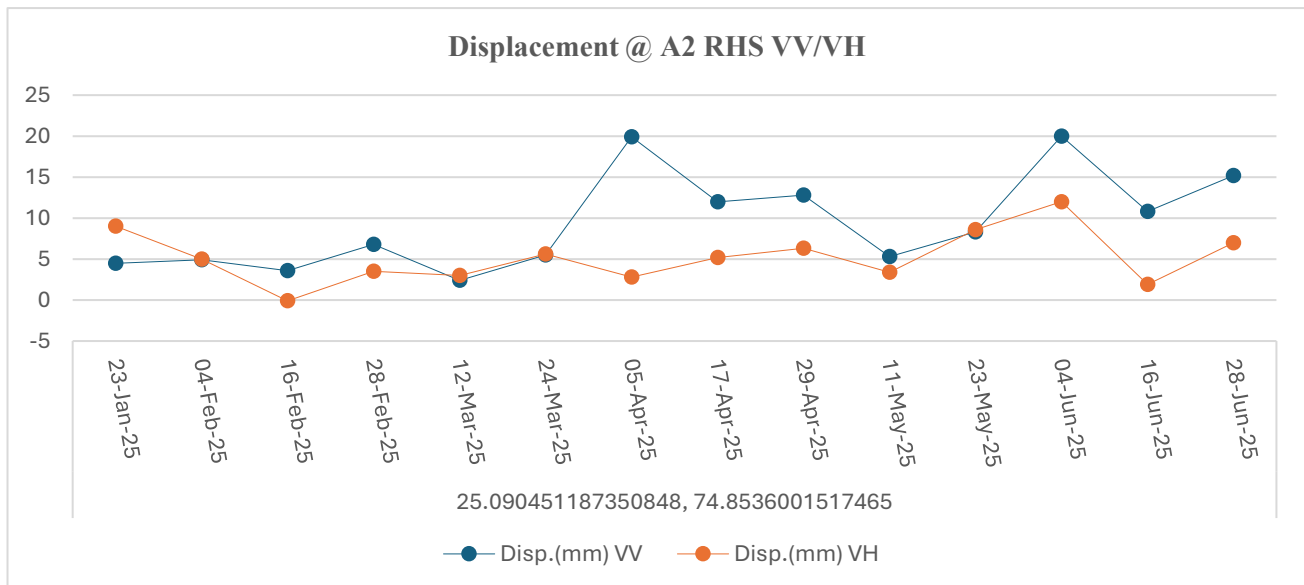
Critical Events & Anomalies:

- **Feb 2025:** VV experiences an early significant negative drop (approx. –10 mm).
- **March 2025:** Another sharp VV decline occurs, reaching roughly 12 mm, followed by a slow recovery trend.
- **June 2025:** VV remains negative (around –15 mm) while VH returns to moderate positive values (around +4 mm).

3.10 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.090451187350848, 74.8536001517465)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** shows moderate-to-high variability, generally ranging from 0 mm up to 20 mm, with several prominent upward peaks (notably in early April and early June). The VV values exhibit a rising trend in the later months.
- **Horizontal Displacement (VH)** remains mostly positive and stable, with readings between 0 mm and +10 mm. VH shows a mild increasing trend through April–June, indicating consistent lateral movement but without instability. Component.

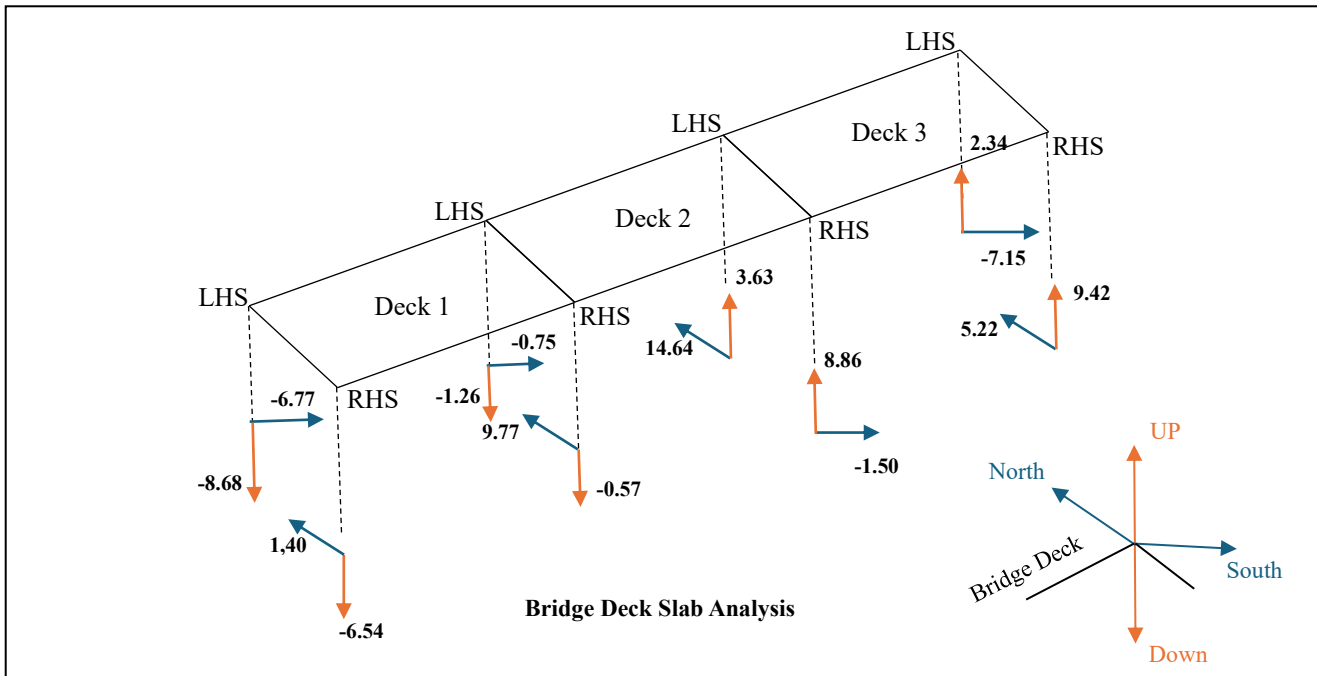
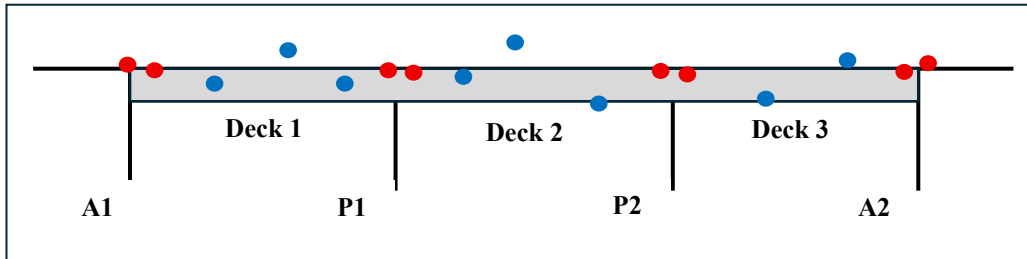
Critical Events & Anomalies:

- **Feb 2025:** A mild early dip in both VV and VH occurs (VV near –2 mm, VH around –3 mm).
- **May 2025:** VH reaches one of its higher values (around +10 mm), corresponding with a moderate VV rise.
- **June 2025:** VV peaks again (+20 mm), followed by a minor drop but remains elevated overall.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The vertical displacement values across Decks 1, 2, and 3 exhibit a combination of upward (positive) and downward (negative) movements at both LHS and RHS locations. Overall, the magnitudes fall within expected serviceability ranges for deck slab systems and do not suggest any excessive deformation.

Horizontal Displacement Behaviour:

- Horizontal displacements across the three decks show a mix of northward and southward shifts, all within acceptable engineering tolerance. The horizontal movements appear uniform across all measurement points and do not indicate any distortion or differential shifting between deck segments.

CONCLUSION:

- The combined vertical and horizontal displacement data for Decks 1, 2, and 3 confirm that the bridge deck slab system is behaving normally under environmental, thermal, and traffic-induced loading. The overall behaviour is consistent with a healthy and serviceable bridge deck system.